Q.27 Explain the block diagram of an optical fiber communication system. (CO1)

Q.28 Explain different types of optical fibers based on refractive index of core. (CO2)

Q.29 Write a short note on absorption losses in optical fiber. (CO3)

Q.30 Write a short note on Population Inversion. (CO4)

Q.31 Write the 5 characteristics of SOA. (CO6)

Q.32 Write any 5 advantages of EDFA. (CO6)

Q.33 Write any 5 applications of LASER. (CO4)

Q.34 Explain the working principle of PIN diode with the help of diagram. (CO5)

Q.35 Classify the mode of propagation in optical fiber. (CO2)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 What is LED? Explain different types of Led in detail. (CO4)

Q.37 Explain the construction and working principle of APD Diode. (CO5)

Q.38 Draw and explain block diagram of semiconductor optical amplifier. (CO6)

(**Note:** Course outcome/CO is for office use only)

No. of Printed Pages : 4 181054/171054/121054 /031054B

Roll No. ....

5th Sem / Eltx, Power Eltx
Subject:- Optical Fiber Communication

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 A ray of light will undergo total internal reflection in it. (CO1)

a) Goes from rarer medium to denser medium
b) Incident at an angle less than the critical angle
c) Strikes the interface normally
d) Incident at an angle greater than the critical angle

Q.2 Multimode step index fiber has a large core diameter of range is______ (CO2)

a) 100 to 300 μm b) 100 to 300 nm
c) 200 to 500 μm d) 200 to 500 nm

Q.3 Single mode fibers allow single mode propagation; the cladding diameter must be at least______ (CO2)

a) Twice the core diameter
b) Thrice the core diameter
c) Five times the core diameter
d) Ten times the core diameter

Q.4 The attenuation losses are measured in terms of (CO3)

a) bel b) decibel
c) del d) None

Q.5 Numerical aperture is constant in case of step index fiber. (CO2)
a) True b) False

Q.6 Optical fibers for communication use are mostly fabricated from______ (CO1)
a) Plastic
b) Silica or multi component glass
c) Ceramics
d) Copper

Q.7 How many types of sources of optical light are available? (CO4)
a) One b) Two
c) Three d) Four

Q.8 The minimum attenuation level which can be achieved in optical fiber communication is (CO1)
a) 4db/km b) 6db/km
c) 0.4 db/km d) 0.2db/km

Q.9 _______ converts the received optical signal into an electrical signal. (CO5)
a) Detector b) Attenuator
c) Laser d) LED

Q.10 The more advantages optical amplifier is________ (CO6)
a) Fiber amplifier
b) Semiconductor amplifier
c) Repeaters
d) Mode hooping amplifier

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 List one application of optical fiber communication. (CO1)

Q.12 Give the formula of Snell's law of refraction. (CO1)

Q.13 LASER is a coherent source of light.(T/F) (CO4)

Q.14 A permanent joint between the individual optical fibers are known as______ (CO2)

Q.15 Define dispersion of light. (CO3)

Q.16 The responsivity of a photodetector is defined as the ratio of the output current or voltage signal to the power of the input optical signal.(T/F) (CO5)

Q.17 Expand EDFA. (CO6)

Q.18 What is Full form of LASER. (CO4)

Q.19 Expand EDFA. (CO5)

Q.20 A single mode fiber has low intermodal dispersion than multimode fiber.(T/F) (CO2)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 List any 5 advantages of fiber optic communication. (CO1)

Q.22 What is total internal reflection? (CO1)

Q.23 Explain scattering losses in optical fiber. (CO3)

Q.24 Write a short note on splice. (CO2)

Q.25 Write a short note on MIE scattering. (CO3)

Q.26 Write any 5 characteristics of Laser. (CO4)